

Dashboard Design

Date	24 September 2024
Team ID	PNT2022TMIDxxxxxx
Project Name	Title of Project
Maximum Marks	5 Marks

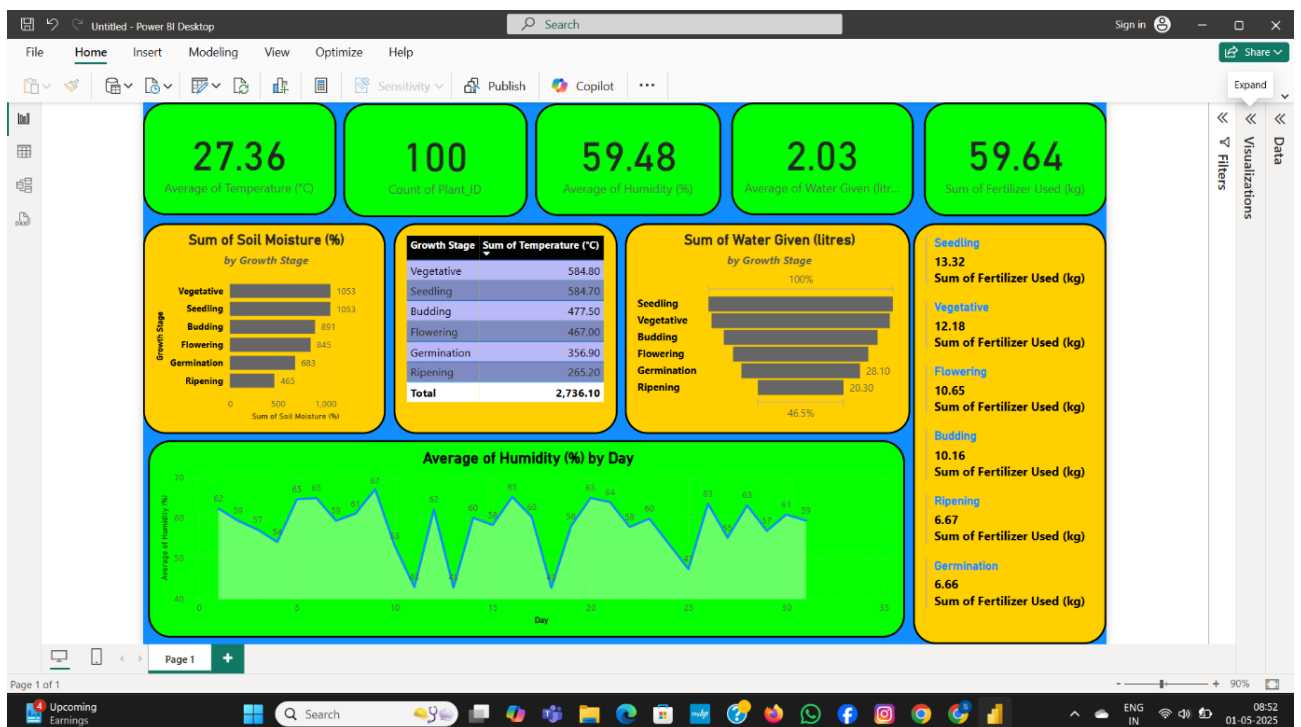
Creating an effective dashboard involves thoughtful design to ensure that the presented information is clear, relevant, and easily understandable for the intended audience. Here are some key principles and best practices for dashboard design

Activity 1: Interactive and visually appealing dashboards

To design a dashboard that is engaging and insightful, the following best practices were implemented:

- **Clear and Intuitive Layout:** Visuals are arranged logically with proper grouping (environmental insights on one side, growth stages on the other).
- **Use of Appropriate Visualizations:** Line charts, bar charts, pie charts, scatter plots, and combo charts were selected based on the type of data being analyzed.
- **Colour and Theming:** A consistent theme with high-medium-low color gradients (e.g., blue for high humidity, orange for low) enhances clarity.

- **Interactive Filters and Slicers:** Slicers were added for filtering data by growth stage and date for targeted analysis.
- **Drill-Down Capabilities:** Enabled on visuals like humidity trend and temperature analysis to explore detailed daily trends.
- **Responsive Design:** Visuals auto-adjust based on screen resolution and layout structure.
- **Custom Visuals and Icons:** Infographics and icons were used to visually represent growth stages.
- **Use of Infographics:** Symbols for fertilizer, water, and plant stages were used to make the dashboard visually engaging.



➤ Highlighted Dashboard Outcomes

- **Growth Stage Distribution:** A pie chart shows that "Vegetative" and "Ripening" stages have the highest record counts, indicating key development periods.
- **Average Temperature by Growth Stage:** A line chart reveals that temperature increases consistently until the flowering stage, then drops slightly in ripening.
- **Water and Fertilizer Usage Analysis:** Bar charts help identify that the "Seedling" and "Vegetative" stages require the most water and fertilizer combined.
- **Humidity Trends Over Time:** A date-wise line graph shows fluctuations, with peaks aligning to growth-sensitive stages.
- **Environmental Factor Comparison:** A combo chart comparing temperature, humidity, and soil moisture shows clear correlations with different growth stages.
- **Fertilizer vs Water Correlation:** A scatter plot with a trendline indicates a moderate positive relationship between water and fertilizer usage.

- **Stage-Based Recommendations:** Based on the interactive dashboard, actionable insights are derived for each growth stage for optimal input planning.

Dashboard Filtering: Filters allow analysis of data by specific date ranges and growth stages, enabling targeted planning.